AI-ML Assignment – 10

Total Marks: 10

Topic

End-to-End Machine Learning Model Deployment using GitHub and Render

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📅 Submission Deadline

01 August 2026, 11:59 PM IST

📄 Assignment PDF

Download the assignment from the shared Google Drive folder:

Google Drive Folder

https://drive.google.com/drive/folders/14_OGcBqYcEHc0F8Zfkezeqgf7LFcoDgF?usp=drive_link

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📊 Dataset

Heart Disease Prediction Dataset

Kaggle Link:

https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset

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Problem Statement

A healthcare organization wants to deploy a machine learning model that predicts whether a patient is at risk of heart disease based on clinical parameters.

Develop a machine learning model, create a REST API using **Flask**, publish the project on **GitHub**, and deploy it as a live web service using **Render**.

Assignment Tasks (Total: 10 Marks)

Task 1: Data Understanding and Preprocessing (2 Marks)

Perform the following:

1. Load the dataset using Pandas.
2. Display the first five records.
3. Identify:
   - Numerical features
   - Target variable
4. Check for missing values.
5. Split the dataset into **80% training** and **20% testing**.

Task 2: Model Development (2 Marks)

Build a classification model using any **one** of the following algorithms:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

Train the model and evaluate it using:

- Accuracy Score

Save the trained model using **Joblib** or **Pickle**.

Task 3: API Development (2 Marks)

Develop a REST API using **Flask**.

The API should:

- Load the trained model.
- Accept patient details as JSON input.
- Return the prediction as JSON.

Example Response

```
{
    "prediction": "Heart Disease Detected"
}
```

Task 4: GitHub and Cloud Deployment (3 Marks)

Complete the following:

GitHub

Create a **public GitHub repository** containing:

- Source Code
- Trained Model
- app.py
- requirements.txt
- README.md

Render Deployment

Deploy the Flask application using **Render**.

The deployed application must:

- Be publicly accessible.
- Successfully return predictions.
- Remain active during evaluation.

Include the deployed application URL in the README.

Task 5: Conclusion (1 Mark)

Write a **100–150 word** conclusion covering:

- Model performance.
- Challenges faced during deployment.
- Importance of MLOps in machine learning projects.

Repository Structure

Your GitHub repository should follow the structure below:

```
HeartDiseaseDeployment/
│
├── app.py
├── model.pkl
├── requirements.txt
├── README.md
├── train_model.py
├── heart.csv
├── templates/
│   └── index.html (Optional)
└── static/ (Optional)
```

Submission Instructions

Create a **public GitHub repository** containing:

- Complete source code
- Trained model
- Deployment files
- README
- Render Deployment URL

Google Form Submission

Submit your assignment using the Google Form:

https://forms.gle/fFL2CFooc5Vb2MXq8

Fill in:

- Name
- Registration Number
- Application Number
- Batch Number
- **Assignment Number (Assignment -10)**
- Public GitHub Repository Link
- **Render Deployment URL**
- Email Address

Important Instructions

- **Deadline: 03 August 2026, 11:59 PM IST**
- Only the **first valid submission** will be accepted.
- Late submissions will be automatically rejected.
- Duplicate submissions will be automatically rejected.
- Future assignment submissions are not permitted.
- Ensure your GitHub repository remains **public** until evaluation is completed.
- Ensure your **Render deployment is active and accessible** during evaluation.
- You will receive an email confirming whether your submission has been **accepted** or **rejected**.

Learning Outcome

After completing this assignment, students will be able to:

- Build and evaluate a machine learning classification model.
- Save and load trained models using **Joblib** or **Pickle**.
- Develop a REST API using **Flask**.
- Manage project code using **GitHub**.
- Deploy machine learning applications on the cloud using **Render**.
- Understand the fundamentals of **MLOps**, including model packaging, version control, deployment, and serving predictions through an API.

Evaluation Rubric (10 Marks)

Criteria	Marks
Data preprocessing and model training	2
Model evaluation and serialization	2
Flask API implementation	2
GitHub repository + Render deployment	3
Documentation and conclusion	1